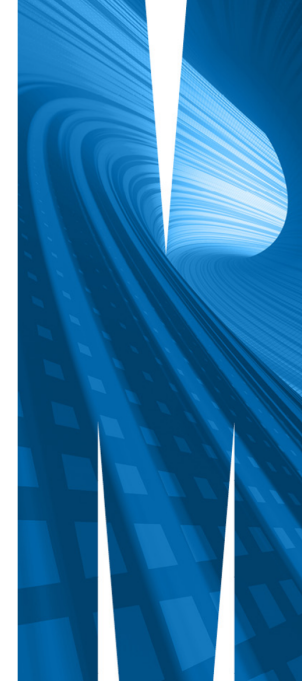
MONASH University

ETX1100/ETX5900 Business Statistics

Week 1: Describing and visualising data

24 November 2026



Outline

1. Unit orientation
2. From business question to data
3. Excel as the analysis workspace
4. Describe categorical data
5. Describe numerical data

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Welcome to Business Statistics

- Business statistics turns data into evidence for decisions.
- This unit combines statistical reasoning, business context and Excel.
- We will use the same workflow throughout:

Business question $\rightarrow$ **method** $\rightarrow$ **Excel** $\rightarrow$ **validation** $\rightarrow$ **plain-language conclusion**

What will we learn?

Across the eight weeks, you will learn to:

- classify and visualise business data;
- calculate and interpret probabilities;
- reason from samples and test business claims;
- analyse relationships between variables;
- build simple and multiple regression models; and
- formulate and solve business modelling problems in Excel.

The eight-week pathway: first half

- **Week 1 · 24 Nov 2026:** From business question to an effective data story
- **Week 2 · 01 Dec 2026:** Probability and distributions
- **Week 3 · 08 Dec 2026:** Sampling and estimation: confidence intervals for means and proportions
- **Week 4 · 15 Dec 2026:** Testing business claims: hypothesis tests for means and proportions

The eight-week pathway: second half

- **Week 5 · 05 Jan 2027:** Categorical independence, chi-square testing and simple linear regression
- **Week 6 · 12 Jan 2027:** Multiple regression and model evaluation
- **Week 7 · 19 Jan 2027:** Spreadsheet business modelling with the Analysis ToolPak
- **Week 8 · 29 Jan 2027:** Integrated business case, communication and revision

How each three-hour class works

- A business question motivates the lecture explanation.
- **Watch me:** follow a short narrated Excel demonstration.
- **Now you try:** apply the same idea to a related task.
- Compare, debug and explain the result before the next concept.
- Finish by stating what the evidence supports and where it is limited.

Explanation and supervised Excel practice are integrated in one class. Bring desktop Excel and your own working copy of the weekly workbook.

Assessment at a glance

- **Weekly classroom exercises — 15%:** group work, due at the end of each class.
- **Weekly quizzes — 15%:** individual; best 5 of 8 count, due Sundays at 17:00.
- **Written assignment — 20%:** individual; due Mon 04 Jan 2027 at 11:55 pm. It assesses material available by the end of Week 4.
- **Final examination — 50%:** individual; held in the examination period and covers the full unit.

Check Moodle for submission and examination arrangements.

Where to find the materials

- The course website has the weekly schedule, slide decks and workbooks.
- Moodle carries official announcements, quizzes and submission details.
- Keep an editable Excel working copy; the posted workbook remains your starting point.
- Check the current unit information when planning your assessment work.

Today’s question

How can a business turn a worksheet of observations into a defensible data story?

1. Define the question, population and variables.
2. Choose a summary and chart for each variable.
3. Build the summary in Excel and check its denominator and units.
4. Explain what the evidence shows and what it cannot show.

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What is business statistics?

ETF defines business statistics as **the application of statistical tools and methods to analyse data for evidence-based business decisions**.

A business question comes first. For example: *Do property prices differ between homes inside and outside a school zone?* The answer needs a clear unit, a relevant population, useful measurements and a conclusion with limitations.

Variable, observation and data

- A **variable** is a characteristic of an item or individual.
- **Data** are the observed values of variables.
- An **observation** is one item or individual measured.

In the `RealEstate` sheet of the Week 1 workbook, one row is one property. `Price ($000s)`, `Beds`, `SchoolZone` and `Auction` are variables. A price of 1100 means AUD 1.1 million.

Population, sample, parameter, statistic

- **Population:** all members of the group about which we want a conclusion.
- **Sample:** a subset selected for analysis, ideally representative of that population.
- **Parameter:** a numerical description of the population, usually unknown.
- **Statistic:** a numerical description computed from the sample.

A mean calculated from 329 sampled properties is a **statistic**. It is evidence about a population mean only if the sample and population are defined and the sampling process is credible.

Descriptive and inferential statistics

What we do	Purpose	Example
Describe	Collect, summarise and present observed data	Report the sample median price and a box plot
Infer	Use sample data to draw a conclusion about a population	Estimate the population mean price

Today we build the description. Weeks 3–4 use sampling to quantify the uncertainty in inference.

Classify the variable before choosing a chart

Type	Meaning	Example	First chart choice
Nominal categorical	Labels without an order	`Auction` Yes/No	Bar chart
Ordinal categorical	Labels with an order	Poor/Fair/Good	Ordered bar chart
Discrete numerical	A count	`Beds`	Bar chart
Continuous numerical	A measurement	`Price ($000s)`	Histogram or box plot

Watch me: classify the RealEstate sheet

Open `RealEstate`. Ask for each column:

1. What does one row represent?
2. Is the entry a label, count or measurement?
3. Which question could a chart answer?
4. Is a number an identifier, a category code or a true quantity?

`Beds` is a count even though Excel stores it as a number. `SchoolZone` is categorical even if we later encode Yes and No as 1 and 0.

Now you try: classify Stores

Time allowed: 5 minutes

Open `Stores`. Identify the observational unit, the type of `Sales`, `Profit` and `Open 24 Hrs?`, and one sensible chart for each. State a possible population before you discuss inference.

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Cells, formulas and ranges

ETF's Excel introduction starts with three rules:

- A cell address combines a column letter and row number: `B2`.
- A formula begins with =; for example, `=C2/B2`.
- A colon identifies a consecutive range: `B2:B76`.

Functions such as `=AVERAGE(B2:B76)`, `=COUNTIF(D2:D76,"Yes")` and `=STDEV.S(B2:B76)` calculate summaries. Check that each range contains the intended records.

Watch me: make an auditable rate

In `Stores`, `Sales` is column B and `Profit` is column C. Put **Profit margin** in a blank column and enter `=C2/B2` for the first store. Format as a percentage, fill down and inspect the last row.

A rate has a numerator, denominator and meaning. `Profit/Sales` answers a different question from profit in dollars. Keep the original data and the formula visible.

Now you try: compare stores fairly

Time allowed: 5 minutes

Calculate the profit margin for every store. Identify a store with high profit in dollars and a store with high profit margin. Write one sentence explaining why the rankings can differ.

Filters and PivotTables

- **Data > Filter** isolates records without deleting other observations.
- **Insert > PivotTable** counts or summarises groups from the full table.
- `COUNTIF` and `COUNTIFS` provide an independent check of a count.

The Excel work should follow the question: choose a numerator, choose the right denominator, then calculate and label the result.

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Frequency and relative frequency

A **frequency** is a count. A **relative frequency** is that count divided by the relevant total.

For a chart of `Auction` by Yes/No, each bar can show either a count or a percentage of all sampled properties. For a comparison **within** school-zone groups, divide by each group's number of properties. State the denominator in the title or note.

Watch me: PivotTable to bar chart

ETF Seminar Week 2 demonstrates this sequence with Real Estate data:

1. Select the `RealEstate` table and choose **Insert > PivotTable**.
2. Drag `Auction` to **Rows** and **Values**; summarise Values by **Count**.
3. For a percentage, choose **Show Values As > % of Grand Total**.
4. Select the summary and choose **Insert > Column or Bar Chart**.
5. Add a descriptive title, a percentage axis label and the

Now you try: create and explain a rate chart

Time allowed: 8 minutes

Use a PivotTable to compare the **within-group Auction rate** for `SchoolZone` = Yes and No. Make a labelled bar chart in Excel. Audit one bar with `COUNTIFS` divided by the correct group total, then explain the comparison without claiming causation.

Visual integrity is part of the result

The ETF visualisation lecture asks whether a nonexpert can understand the graph. Check:

- Purpose, informative title and readable labels.
- Units and denominator.
- Comparable axis scale; start a bar chart at zero.
- A legend when there is more than one series.
- No distorted sizes or selective categories.

A clear chart still needs a sentence stating its main pattern and a limitation.

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One number cannot describe a distribution

Numerical description has three parts:

1. **Centre:** mean and median describe a typical value.
2. **Spread:** standard deviation and IQR describe variability.
3. **Shape:** symmetry, skewness and unusual values affect interpretation.

ETF Seminar Week 4 uses all three when comparing house prices by school zone. A chart shows features a single average can hide.

Mean, median and mode

- **Mean:** sum of observations divided by their number, `=AVERAGE(range)`. It uses every value and is affected by extreme prices.
- **Median:** the middle ordered value, `=MEDIAN(range)`. Half the values are at or below it.
- **Mode:** the most frequent value, `=MODE.SNGL(range)`. For prices with few repeats, a modal interval in a histogram is often more useful.

If a few very expensive houses pull the mean above the median, describe the distribution as right-skewed before choosing the headline statistic.

Spread: standard deviation and IQR

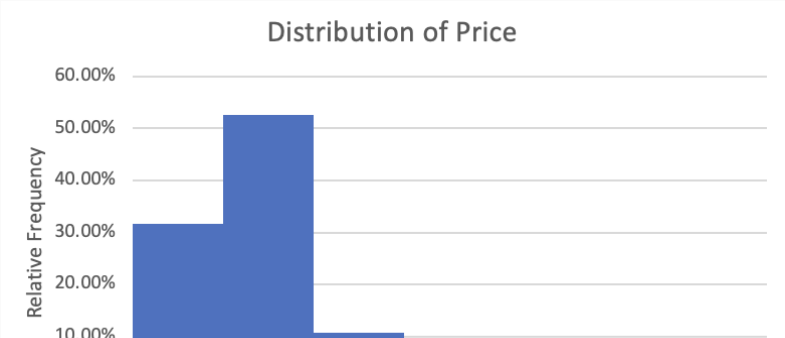
The **sample standard deviation** measures how observations vary around the sample mean: `=STDEV.S(range)`. It uses all values but responds to outliers.

The **interquartile range** is `IQR = Q3 - Q1`, the spread of the middle 50%. Use `=QUARTILE.EXC(range,1)` and `=QUARTILE.EXC(range,3)` for the ETF convention, and label that choice. Excel also has `QUARTILE.INC`; use one convention consistently.

The range, `MAX - MIN`, relies only on the two extremes. Give units for every spread statistic.

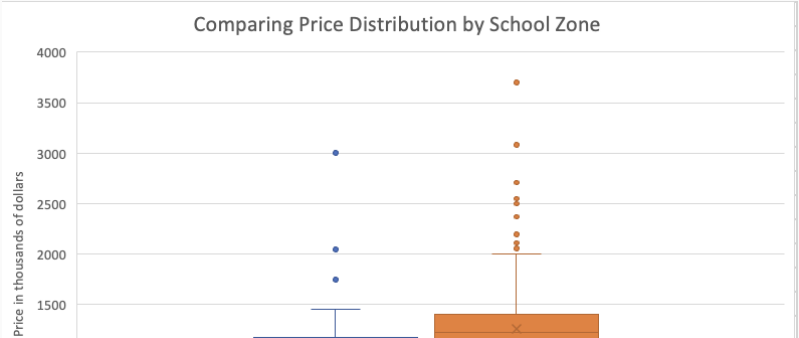
What a histogram shows

A histogram groups a continuous variable into adjacent intervals. Its bars touch, and their pattern can reveal skewness, a long tail or unusual values. ETF groups Real Estate prices in AUD 500,000 bins, then sets the chart's gap width to zero.



What a box plot shows

The box spans Q1 to Q3, with the median inside. Whiskers and separate points show the range and unusual values under Excel's box-plot rule. A side-by-side box plot compares centre, spread and outliers on one scale.



Watch me: group summaries and a box plot

In `RealEstate`, compare `Price ($000s)` for `SchoolZone` Yes and No. ETF Workshop Week 4 uses **Data > Data Analysis > Descriptive Statistics** for mean, median and sample standard deviation. Add Q1 and Q3 with formulas.

For the chart, put the two price groups in adjacent columns with clear headers, then choose **Insert > Statistical Chart > Box & Whisker**. Label the vertical axis **Price (AUD thousands)**. The ToolPak provides summaries; the ordinary Excel chart displays the distribution.

Now you try: compare two distributions

Time allowed: 10 minutes

Compare price for `New` = Yes and No. Produce a small table of `COUNT`, `AVERAGE`, `MEDIAN`, `STDEV.S`, Q1 and Q3, then make a side-by-side box plot. Which group has the higher typical price? Which has more spread? Identify one visible limitation.

Build a one-slide data story

A business data story has a repeatable order:

Question → unit and population → chart and numerical summary → check → plain-language claim → limitation.

For the school-zone comparison, a higher sample median is a description of the observed properties. It does not by itself show that school-zone status caused the price difference; property size and location may also matter.

Now you try: briefing panel

Time allowed: 8 minutes

Create a compact panel for the Real Estate question with one labelled categorical chart, one numerical distribution chart, two summary numbers and a two-sentence conclusion. Make the Excel working auditable from raw rows to chart.

Optional Excel practice

The `Stores` sheet also supports ETF Workshop Week 1 practice with `VLOOKUP`, `INDEX/MATCH`, `IF`, `IFERROR`, filters, `COUNTIF` and `SUMIF`. Use these when a business question needs a lookup or conditional summary. They are useful tools, but the core Week 1 story is choosing, constructing and interpreting summaries and charts.

Close: what must be true of your chart?

- The chart type fits the variable.
- The groups, counts and denominators are explicit.
- The units, labels and scales are readable.
- The written claim matches the plotted evidence.
- The original rows and formulas remain available for checking.